

Data Visualization

Saryu Onishi

HES 3000-002, Sport Performance Analytics

Why visualization matters in sports analytics

Why visualization matters in sports analytics

- **Pattern recognition and speed**
- Communication and decision support

Who played the most minutes?

Player	Pos	GP	GS	Mins
T. Baribo	F	29	20	1876
B. Damiani	F	33	23	2040
M. Uhre	F	29	16	1325
I. Vassilev	M	31	21	1973
D. Jean Jacques	M	31	26	2361
K. Wagner	D	32	30	2769
Q. Sullivan	M	28	24	1967
N. Harriel	D	25	23	2098
M. Iloski	F	9	7	600
D. Gazdag	M	6	6	498
J. Lukic	M	32	30	2620
O. Makhanya	D	29	23	2145
A. Bedoya	M	24	9	739
J. Glesnes	D	31	30	2626
F. Westfield	D	25	18	1724
I. Glavinovich	D	4	3	257
M. Anderson	F	3	1	98
J. Bueno	M	20	11	903
N. Pierre	D	1	0	20
D. Vazquez	M	1	0	19
O. Mbaizo	D	9	6	481
C. Sullivan	M	11	1	214
J. Rafanello	M	15	2	274
S. Olivas	F	3	0	45
B. Bender	M	7	4	320
A. Rick	G	14	13	1232
E. Davis	F	1	0	2
A. Blake	G	21	21	1828
C. Donovan	F	20	6	540

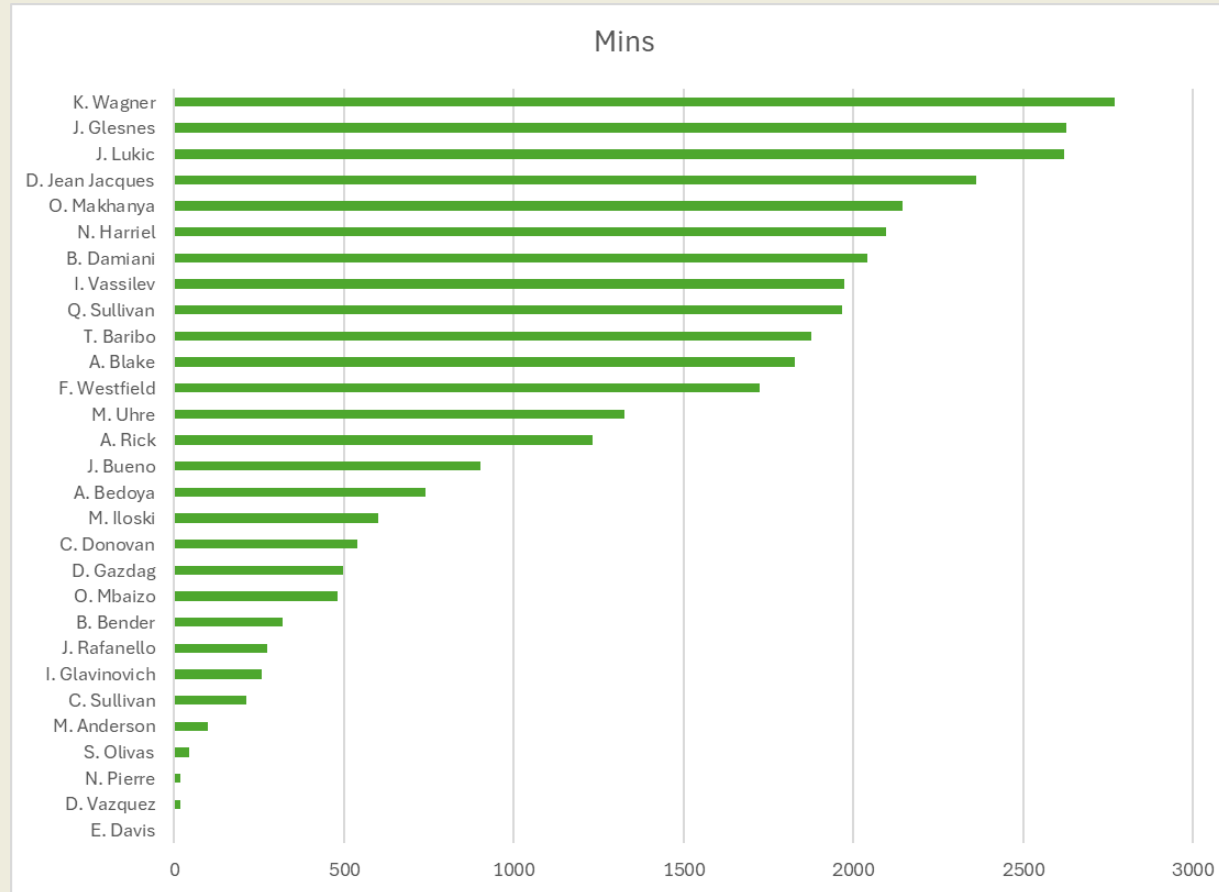
Gradient to represent magnitude

Player	Pos	GP	GS	Mins
T. Baribo	F	29	20	1876
B. Damiani	F	33	23	2040
M. Uhre	F	29	16	1325
I. Vassilev	M	31	21	1973
D. Jean Jacques	M	31	26	2361
K. Wagner	D	32	30	2769
Q. Sullivan	M	28	24	1967
N. Harriel	D	25	23	2098
M. Iloski	F	9	7	600
D. Gazdag	M	6	6	498
J. Lukic	M	32	30	2620
O. Makhanya	D	29	23	2145
A. Bedoya	M	24	9	739
J. Glesnes	D	31	30	2626
F. Westfield	D	25	18	1724
I. Glavinovich	D	4	3	257
M. Anderson	F	3	1	98
J. Bueno	M	20	11	903
N. Pierre	D	1	0	20
D. Vazquez	M	1	0	19
O. Mbaizo	D	9	6	481
C. Sullivan	M	11	1	214
J. Rafanello	M	15	2	274
S. Olivas	F	3	0	45
B. Bender	M	7	4	320
A. Rick	G	14	13	1232
E. Davis	F	1	0	2
A. Blake	G	21	21	1828
C. Donovan	F	20	6	540

Order to organize values

Player	Pos	GP	GS	Mins
K. Wagner	D	32	30	2769
J. Glesnes	D	31	30	2626
J. Lukic	M	32	30	2620
D. Jean Jacques	M	31	26	2361
O. Makhanya	D	29	23	2145
N. Harriel	D	25	23	2098
B. Damiani	F	33	23	2040
I. Vassilev	M	31	21	1973
Q. Sullivan	M	28	24	1967
T. Baribo	F	29	20	1876
A. Blake	G	21	21	1828
F. Westfield	D	25	18	1724
M. Uhre	F	29	16	1325
A. Rick	G	14	13	1232
J. Bueno	M	20	11	903
A. Bedoya	M	24	9	739
M. Iloski	F	9	7	600
C. Donovan	F	20	6	540
D. Gazdag	M	6	6	498
O. Mbaizo	D	9	6	481
B. Bender	M	7	4	320
J. Rafanello	M	15	2	274
I. Glavinovich	D	4	3	257
C. Sullivan	M	11	1	214
M. Anderson	F	3	1	98
S. Olivas	F	3	0	45
N. Pierre	D	1	0	20
D. Vazquez	M	1	0	19
E. Davis	F	1	0	2

Visual representation of values



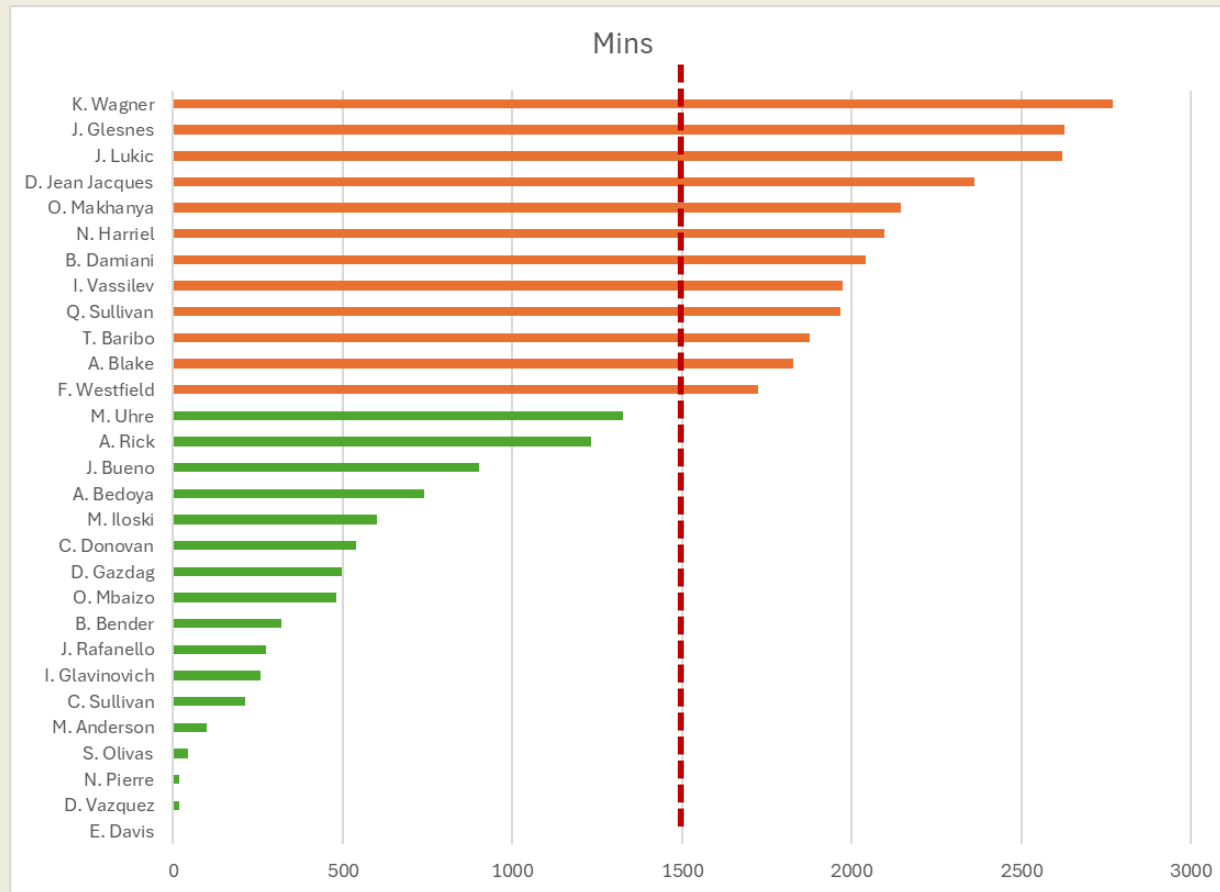
Why visualization matters in sports analytics

- Pattern recognition and speed
- **Communication and decision support**

Who played >1500 minutes last season?

Player	Pos	GP	GS	Mins
T. Baribo	F	29	20	1876
B. Damiani	F	33	23	2040
M. Uhre	F	29	16	1325
I. Vassilev	M	31	21	1973
D. Jean Jacques	M	31	26	2361
K. Wagner	D	32	30	2769
Q. Sullivan	M	28	24	1967
N. Harriel	D	25	23	2098
M. Iloski	F	9	7	600
D. Gazdag	M	6	6	498
J. Lukic	M	32	30	2620
O. Makhanya	D	29	23	2145
A. Bedoya	M	24	9	739
J. Glesnes	D	31	30	2626
F. Westfield	D	25	18	1724
I. Glavinovich	D	4	3	257
M. Anderson	F	3	1	98
J. Bueno	M	20	11	903
N. Pierre	D	1	0	20
D. Vazquez	M	1	0	19
O. Mbaizo	D	9	6	481
C. Sullivan	M	11	1	214
J. Rafanello	M	15	2	274
S. Olivas	F	3	0	45
B. Bender	M	7	4	320
A. Rick	G	14	13	1232
E. Davis	F	1	0	2
A. Blake	G	21	21	1828
C. Donovan	F	20	6	540

Color, order, graphs



Player	Pos	Mins
K. Wagner	D	2769
J. Glesnes	D	2626
J. Lukic	M	2620
D. Jean Jacques	M	2361
O. Makhanya	D	2145
N. Harriel	D	2098
B. Damiani	F	2040
I. Vassilev	M	1973
Q. Sullivan	M	1967
T. Baribo	F	1876
A. Blake	G	1828
F. Westfield	D	1724
M. Uhre	F	1325
A. Rick	G	1232
J. Bueno	M	903
A. Bedoya	M	739
M. Iloski	F	600
C. Donovan	F	540
D. Gazdag	M	498
O. Mbaizo	D	481
B. Bender	M	320
J. Rafanello	M	274
I. Glavinovich	D	257
C. Sullivan	M	214
M. Anderson	F	98
S. Olivas	F	45
N. Pierre	D	20
D. Vazquez	M	19
E. Davis	F	2

Why visualization matters

Each version contained the same basic data. What changed was how the data were visually encoded.

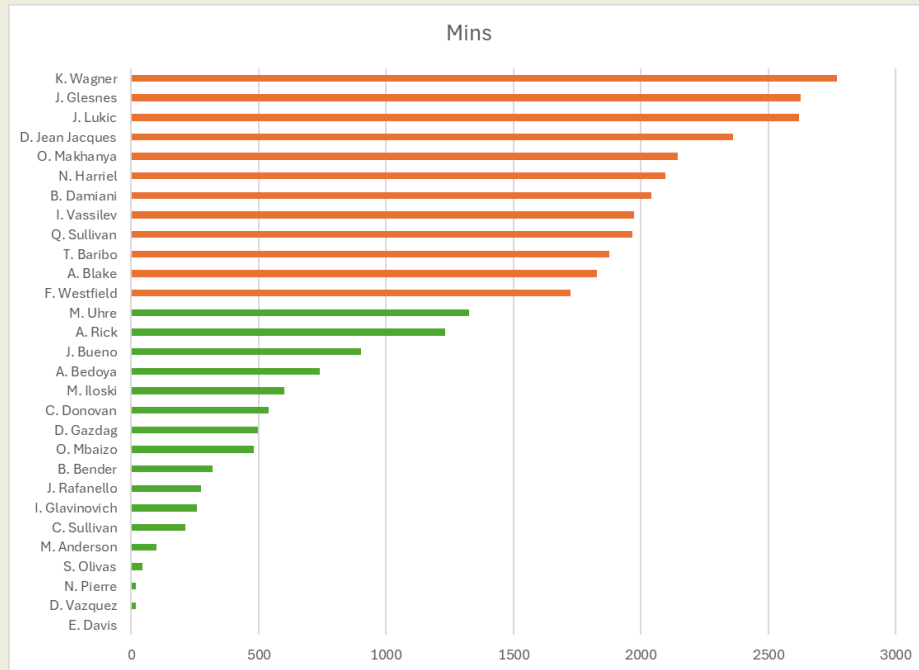
- Tables are useful for looking up exact values.
- Charts are useful for seeing patterns.
- Ordering improves comparison.
- Color and reference lines add meaning.

Different visual choices support different decisions.

Directing attention

Visual encoding

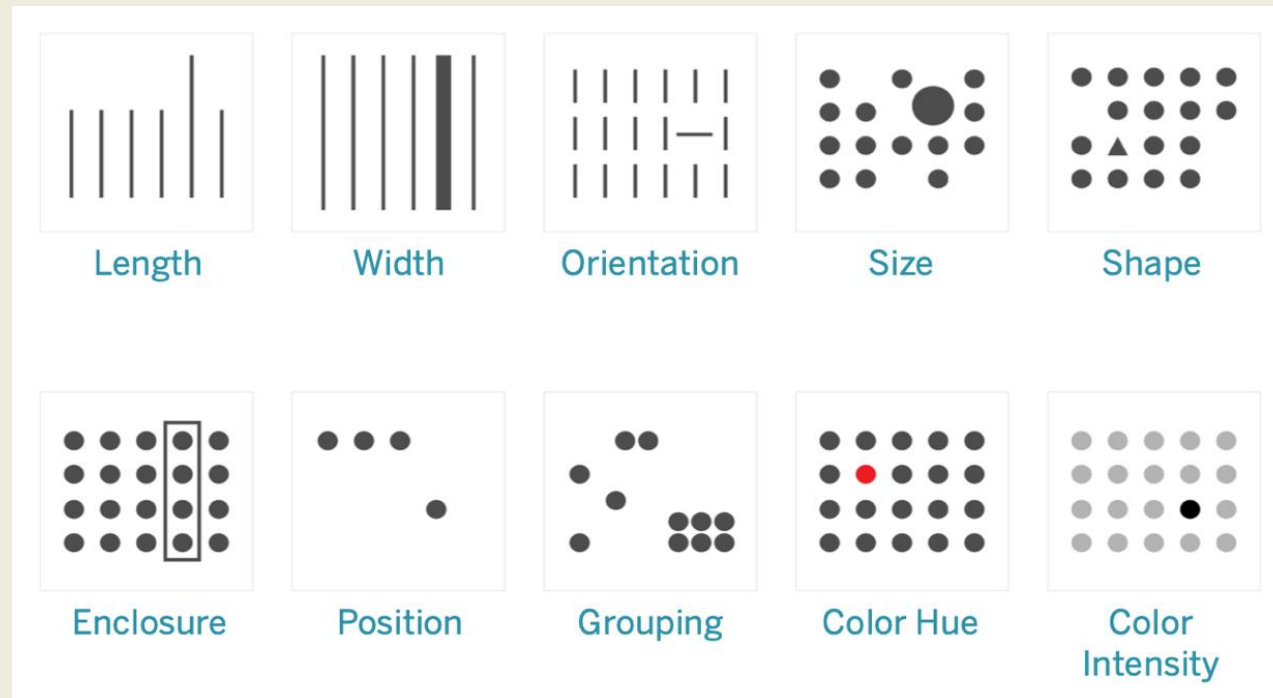
- Visual encoding maps data values to visual properties.
- Attributes such as position, length, color, size and shape communicate different information.



- vertical position = player name
- bar length = minutes
- reference line = decision threshold
- color = above or below threshold

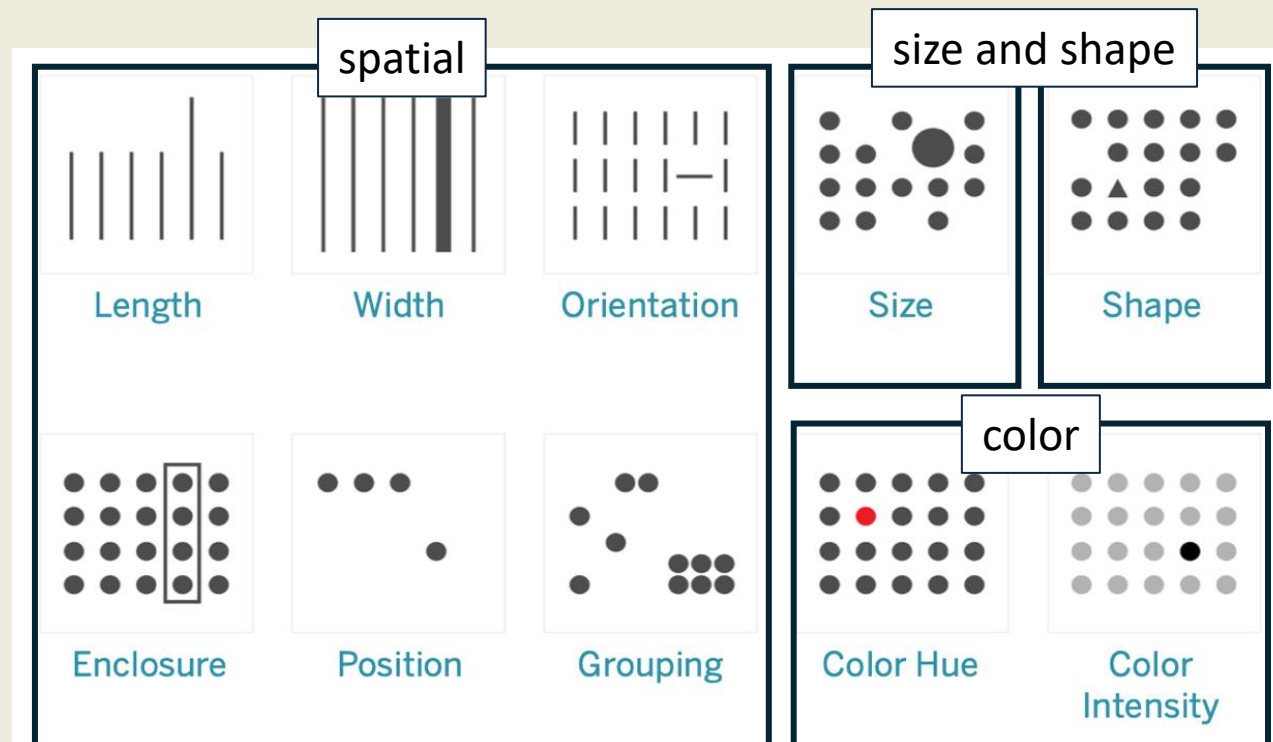
Pre-attentive attributes

- Visual properties that the human brain processes almost instantly and subconsciously



Pre-attentive attributes

- Visual properties that the human brain processes almost instantly and subconsciously



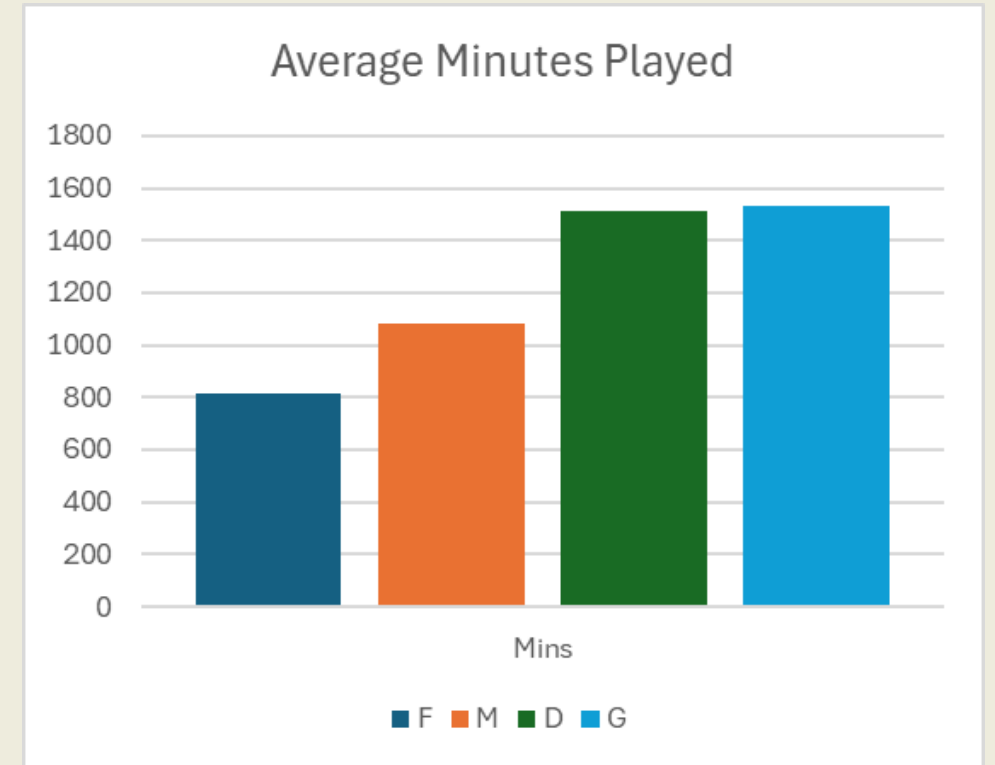
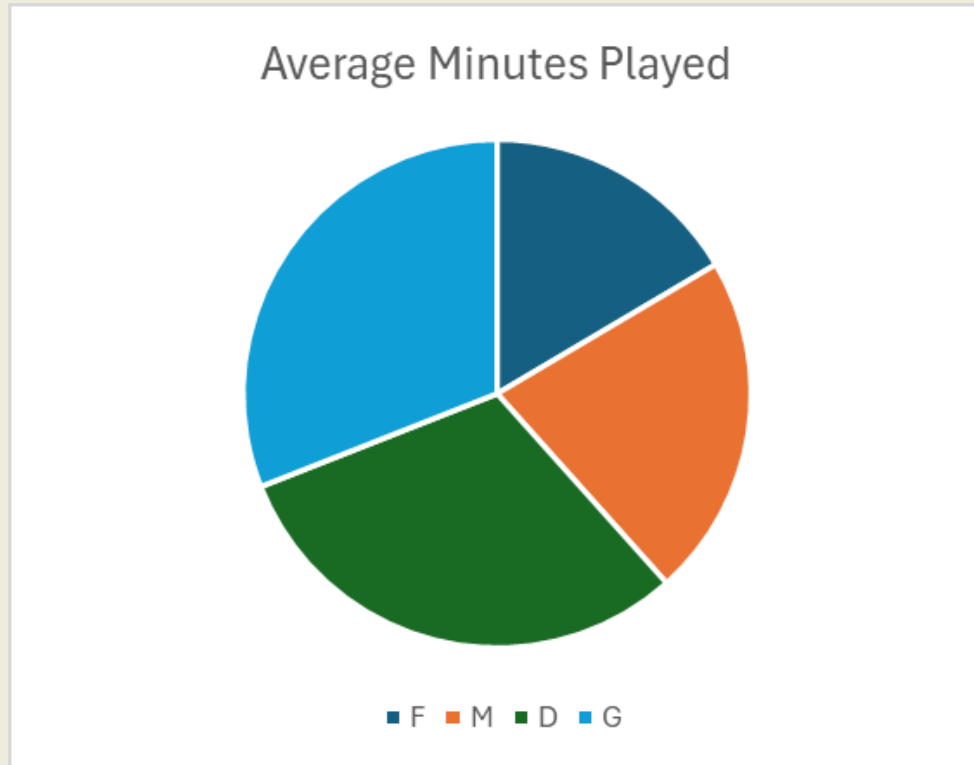
Attention versus importance

- Visual emphasis tells the audience where to look.
- The most visually prominent element appears important.
- Emphasis should reflect the analytical question.
- Unnecessary emphasis can distort interpretation.

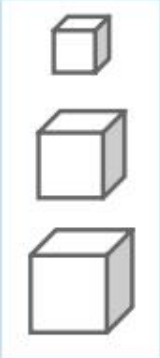
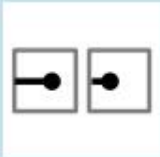
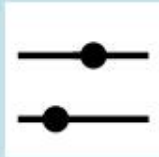
Accuracy in visualizations

Making comparisons easy

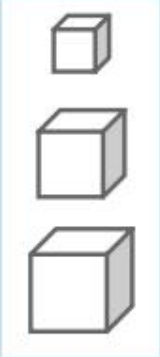
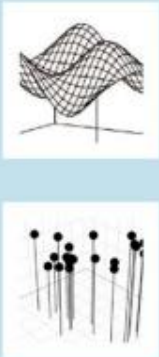
- Which position accumulated more minutes?
- Approximately how many more minutes?



Accuracy of visual encodings

Volume	Color hue	Depth: 3d position	Color intensity	Area	Slope or Angle	Length	Position on unaligned scale	Position on common scale
	 	 	 	 	 	 	 	 

Accuracy of visual encodings

Volume	Color hue	Depth: 3d position	Color intensity	Area	Slope or Angle	Length	Position on unaligned scale	Position on common scale
								

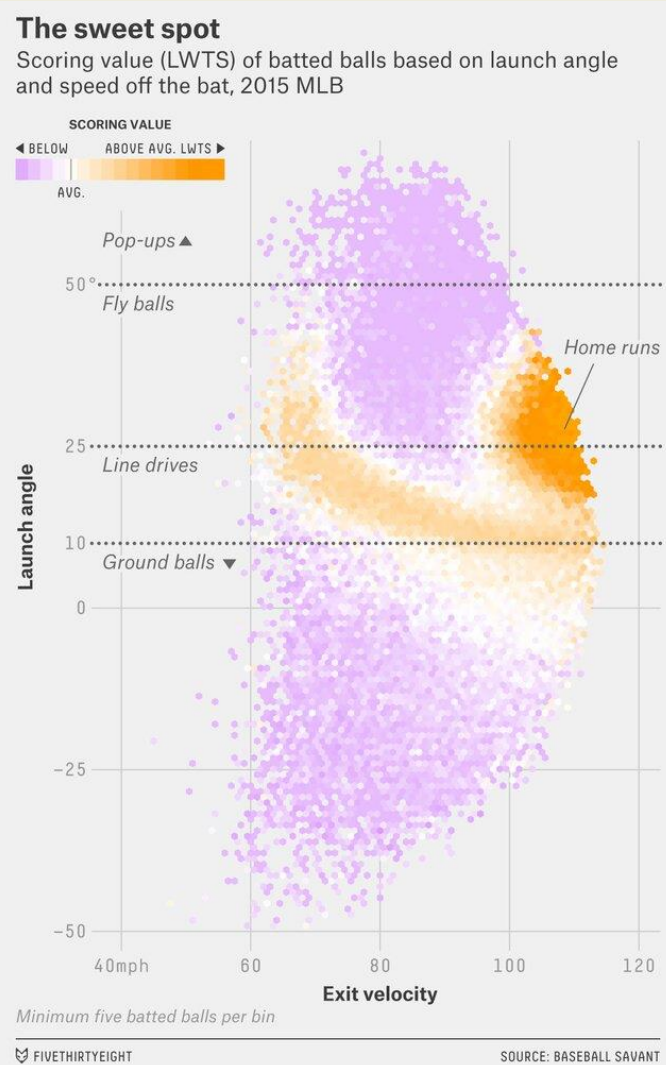
Selecting attributes

Selecting attributes

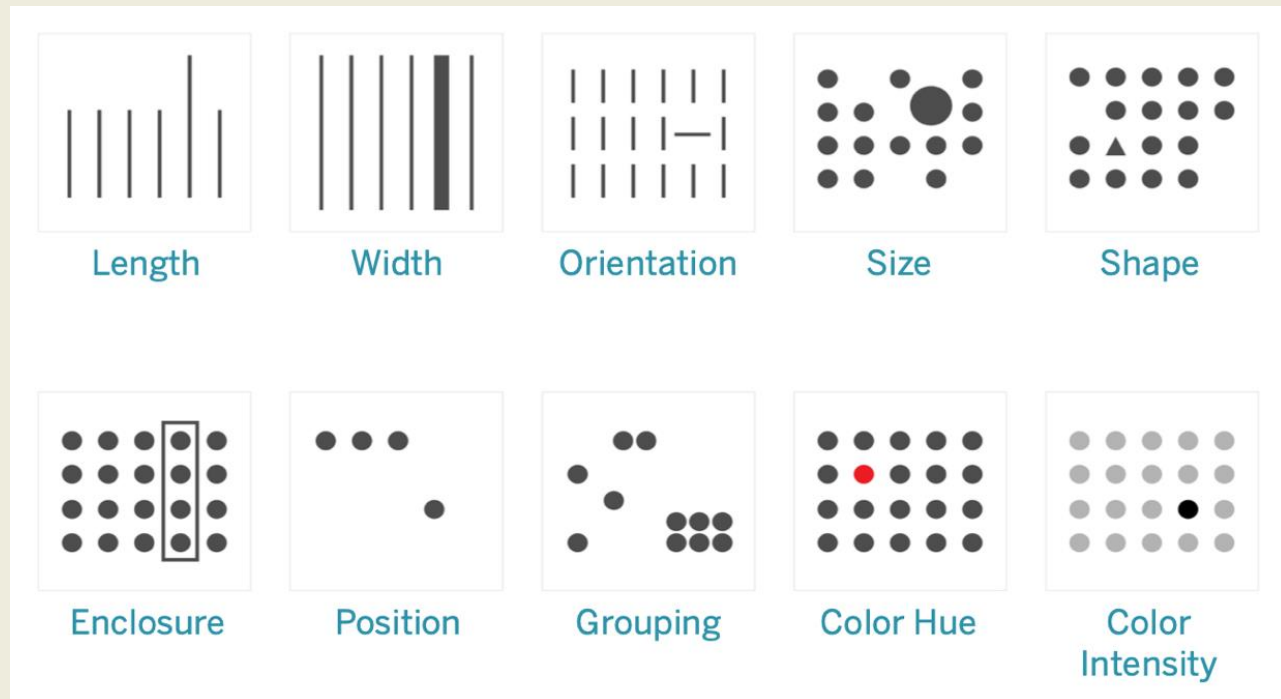
- Which visual properties would appropriately represent:
 - Position
 - Minutes played
 - Team
 - Average sprint speed

Data type	Appropriate visual properties
Numerical	Length, size, color intensity
Nominal	Color hue, shape
Ordinal	Color intensity, size

Identifying visual attributes



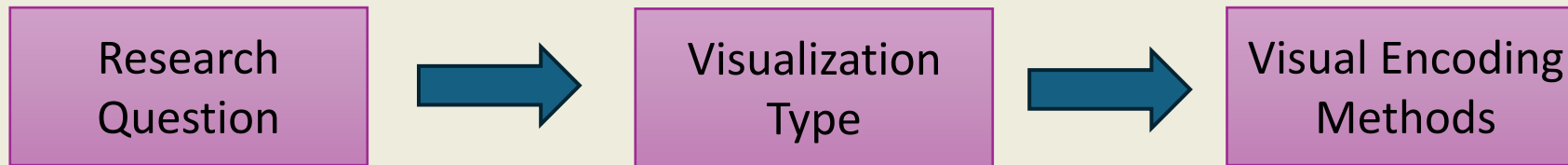
- What visual attributes are used here?



Choosing the Right Chart Type

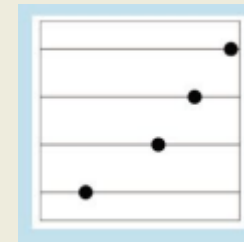
Match the chart to your question

- You would never do analysis before the question
- Same ideas apply here



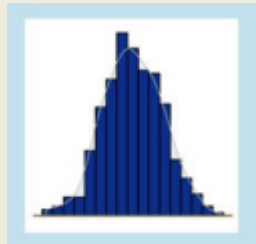
Comparisons (rank)

- Bar charts: Simple stat comparisons (points per game by player)
- Grouped/stacked bars: Multi-category comparisons (shooting % by zone)
- Dot plots: Clean alternative to bars, especially for rankings
- Radar/spider charts: Multi-dimensional player comparisons (use cautiously!)



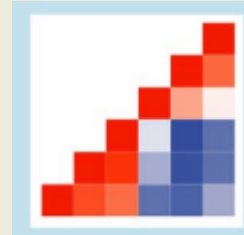
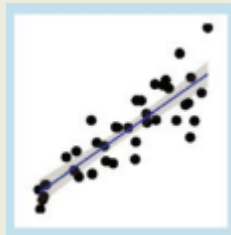
Distribution

- Histograms: Player height distribution, shot distance distribution
- Density plots: Smoothed shooting ranges
- Box plots: Comparing stat distributions across positions or teams
- Violin plots: Showing full distribution shape (sprint speeds by position)



Relationships

- Scatter plots: Exploring correlations (height vs. rebounding)
- Correlation matrices: Multiple pairwise correlation
- Hexbin plots: scatter plots with aggregation (shot charts)



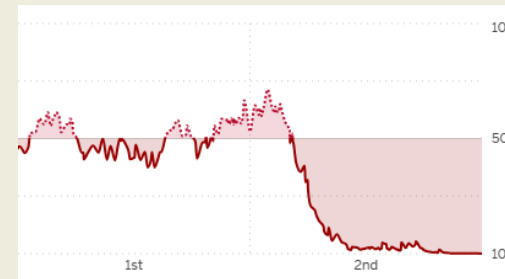
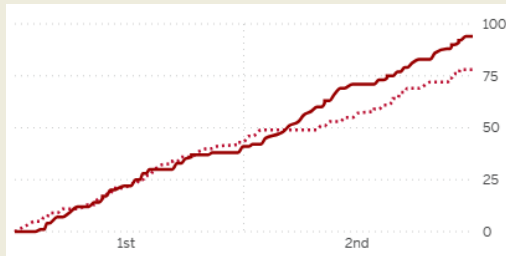
Part-to-whole

- Stacked bars: Composition of team offense (3PT vs. 2PT vs. FT)
- Tree maps: Allocating playing time or usage
- Pie charts: use cautiously



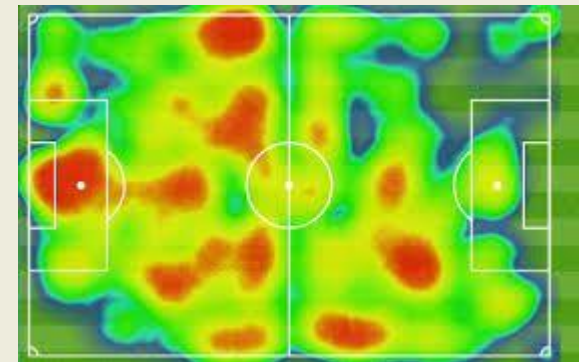
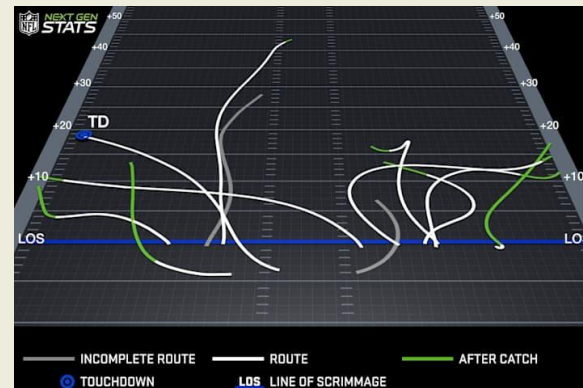
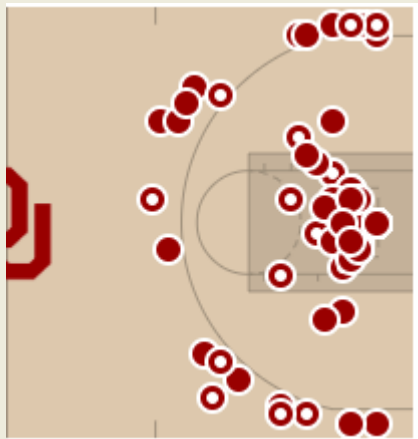
Time series

- Line charts: Performance over season, in-game momentum
- Area charts: Cumulative stats over time



Spatial/Location data

- Heat maps: Player positioning, shot locations
- Contour plots: Density of activity on field/court
- Field/court overlays: Context matters in sports



Bad choices

